Name: _____ score:_____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

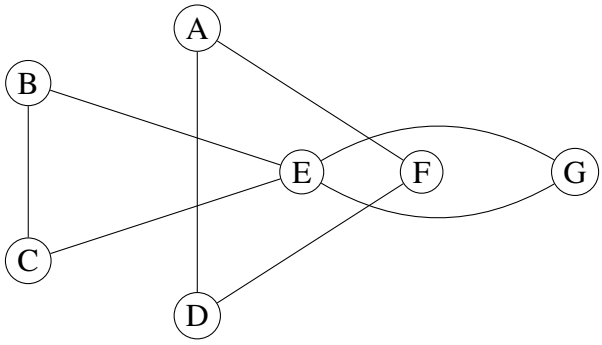
1. (2 points) Describe a situation that can be modeled with a **weighted** graph.

   What do the **vertices** represent?

   What do the **edges** represent?

   What do the **weights** represent?

2. (4 points) Consider the following graph:



   (a) How many vertices does this graph have? _____

   (b) How many edges does this graph have? _____

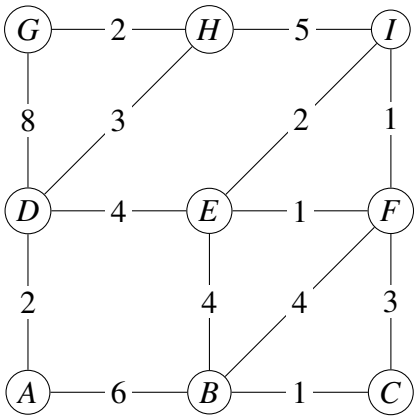
   (c) What is the degree of vertex E? _____

   (d) Explain why this graph is not connected.

3. (4 points) Recall Kruskal's algorithm roughly says the following:

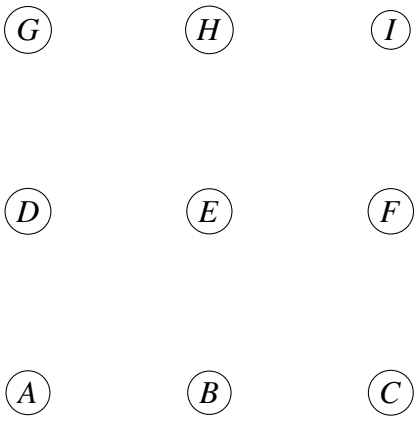
   Kruskal's Algorithm Select the cheapest edge in the graph that does not create a circuit. Stop when a spanning tree is obtained.

   Use Kruskal's Algorithm to find a minimum weight spanning tree in the graph below. Break ties by choosing the alphabetically earliest edge first.



used?	edges	weights
	BC	1
	EF	1
	FI	1
	AD	2
	EI	2
	GH	2
	CF	3
	DH	3
	BE	4
	BF	4
	DE	4
	HI	5
	AB	6
	DG	8

   Draw the minimum weight spanning tree below.



   What is the total weight of the spanning tree you found? _____